



Revision Exercise S3C: Structured Query Language (Solutions)

1(a) Create the Staff table

```
CREATE TABLE Staff (  
    StaffID VARCHAR(5) PRIMARY KEY,  
    Name VARCHAR(50),  
    Gender CHAR(1),  
    DateOfBirth DATE,  
    Address VARCHAR(100),  
    ContactNo VARCHAR(15),  
    Grade VARCHAR(10)  
);
```

1(b) Add Email field to Staff table

```
ALTER TABLE Staff  
ADD Email VARCHAR(50);
```

1(c) Insert a record into the Ward table

```
INSERT INTO Ward (WardNo, Class, NoOfBed, Level, StaffID)  
VALUES ('W05', 'B2', 12, 3, 'S013');
```

1(d) Display Name and ContactNo of all patients

```
SELECT Name, ContactNo  
FROM Patient;
```

1(e) Display staff name (as StaffName) and grade, sorted by grade (descending)

```
SELECT Name AS StaffName, Grade  
FROM Staff  
ORDER BY Grade DESC;
```

1(f) Update contact number of patient P028

```
UPDATE Patient  
SET ContactNo = '91918282'  
WHERE PatientID = 'P028';
```

- 1(g) Delete patients born before 1 January 1980

```
DELETE FROM Patient  
WHERE DateOfBirth < '1980-01-01';
```

- 1(h) Display ward details and name of staff in-charge (Level 2 wards)

```
SELECT Ward.WardNo, Ward.Class, Staff.Name  
FROM Ward  
INNER JOIN Staff ON Ward.StaffID = Staff.StaffID  
WHERE Ward.Level = 2;
```

- 1(i) Display grades with more than 5 staff members

```
SELECT Grade, COUNT(*) AS TotalStaff  
FROM Staff  
GROUP BY Grade  
HAVING COUNT(*) > 5;
```

- 1(j) Display top 3 wards with the highest number of beds

```
SELECT WardNo, NoOfBed  
FROM Ward  
ORDER BY NoOfBed DESC  
LIMIT 3;
```

- 1(k) Remove the Ward table

```
DROP TABLE Ward;
```

2(a) Create ROOMTYPE table

```
CREATE TABLE ROOMTYPE (  
    RoomType VARCHAR(20) PRIMARY KEY,  
    WeekdayRate DECIMAL(8,2),  
    WeekendRate DECIMAL(8,2)  
);
```

2(b) Add SeaView column to ROOM

```
ALTER TABLE ROOM  
ADD SeaView VARCHAR(3);
```

2(c) Insert a record into CUSTOMER

```
INSERT INTO CUSTOMER (CustomerID, Name, Address, Telephone, Email)  
VALUES ('AB025', 'Sarah Tan', '23 Beach Road', '87654321', 'sarah.tan@mail.com');
```

2(d) Display bookings arriving in January 2026

```
SELECT C.Name AS CustomerName, B.RoomNumber, B.ArrivalDate,  
    B.DepartureDate  
FROM BOOKING B  
INNER JOIN CUSTOMER C ON B.CustomerID = C.CustomerID  
WHERE B.ArrivalDate >= '2026-01-01' AND B.ArrivalDate <= '2026-01-31';
```

2(e) Display BookingID, Customer Name, TotalRevenue

```
SELECT B.BookingID, C.Name AS CustomerName,  
    (B.DepartureDate - B.ArrivalDate) * RType.WeekdayRate AS TotalRevenue  
FROM BOOKING B  
INNER JOIN CUSTOMER C ON B.CustomerID = C.CustomerID  
INNER JOIN ROOM R ON B.RoomNumber = R.RoomNumber  
INNER JOIN ROOMTYPE RType ON R.RoomType = RType.RoomType;
```

2(f) Update WeekdayRate for Deluxe rooms

```
UPDATE ROOMTYPE  
SET WeekdayRate = 150  
WHERE RoomType = 'Deluxe';
```

- 2(g) Delete bookings with DepartureDate before 1 Jan 2025

```
DELETE FROM BOOKING  
WHERE DepartureDate < '2026-01-01';
```

- 2(h) Display top 3 customers by total revenue

```
SELECT C.Name AS CustomerName,  
       SUM((B.DepartureDate - B.ArrivalDate) * RType.WeekdayRate) AS TotalSpent  
FROM BOOKING B  
INNER JOIN CUSTOMER C ON B.CustomerID = C.CustomerID  
INNER JOIN ROOM R ON B.RoomNumber = R.RoomNumber  
INNER JOIN ROOMTYPE RType ON R.RoomType = RType.RoomType  
GROUP BY C.Name  
ORDER BY TotalSpent DESC  
LIMIT 3;
```

- 2(i) Number of bookings for sea view rooms in December 2025

```
SELECT COUNT(*) AS NumBookings  
FROM BOOKING B  
INNER JOIN ROOM R ON B.RoomNumber = R.RoomNumber  
WHERE R.SeaView = 'YES'  
AND B.ArrivalDate >= '2025-12-01'  
AND B.ArrivalDate <= '2025-12-31';
```